Ex. No: 12	BUILD WEB APP WITH FLUTTER
Date of Exercise	29.09.2026
Git-Hub Host Link	https://github.com/URK23CS1266/exp12-flutter

Aim:

To develop, run, and build a responsive Flutter application for the Web using Flutter and Chrome.

Description:

This experiment demonstrates how a Flutter allows developers to create web applications using the same Flutter framework used for mobile applications. In this experiment, a Flutter Web application is created and executed in Chrome. A responsive layout is implemented using LayoutBuilder so that the interface changes based on the browser window width. Finally, a production web build is generated using flutter build web.

Procedure:

Step 1: Check the Flutter installation and development environment using: flutter doctor

Step 2: Check the available devices and verify that Chrome or another web browser is available using: flutter devices

Step 3: Create a new Flutter web application using: flutter create flutter_web_app

Step 4: Navigate to the application directory using: cd flutter_web_app

Step 5: Open the Flutter project in VS Code using: code .

Step 6: Create the web interface in lib/main.dart using Scaffold, AppBar, and ElevatedButton.

Step 7: Implement a responsive layout using LayoutBuilder to display different layouts for mobile and desktop screen widths.

Step 8: Run the Flutter web application in Chrome using: flutter run -d chrome

Step 9: Resize the Chrome browser window and verify that the application changes between the Mobile Web Layout and Desktop Web Layout.

Step 10: Create the production web build using: flutter build web and verify that the generated files are available in the build/web/ directory

Program:

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

const Color primaryColor = Color(0xFF6558E8);
const Color backgroundColor = Color(0xFFF6F7FC);

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
    debugShowCheckedModeBanner: false,
    title: 'Flutter Web Application',
      theme: ThemeData(
        useMaterial3: true,
        scaffoldBackgroundColor: backgroundColor,
        colorScheme: ColorScheme.fromSeed(
        seedColor: primaryColor,
        ),
        fontFamily: 'Arial',
      ),
      home: const DashboardPage(),
    );
  }
}

class DashboardPage extends StatelessWidget {
  const DashboardPage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        backgroundColor: Colors.white,
        surfaceTintColor: Colors.white,
        elevation: 0,
        title: const Row(
          mainAxisSize: MainAxisSize.min,
          children: [
            Icon(Icons.bolt, color: primaryColor, size: 30),
            SizedBox(width: 8),
            Text(
              'FlutterSpace',
```

```
          style: TextStyle(
            fontWeight: FontWeight.bold,
            color: Color(0xFF252542),
          ),
        ),
      ],
    ),
    actions: [
      IconButton(
      onPressed: () {},
        icon: const Icon(Icons.notifications_none),
      ),
      const Padding(
        padding: EdgeInsets.only(right: 20),
        child: CircleAvatar(
          backgroundColor: Color(0xFFE8E5FF),
          child: Icon(Icons.person, color: primaryColor),
        ),
      ),
    ],
  ),
  body: LayoutBuilder(
    builder: (context, constraints) {
      final isMobile = constraints.maxWidth < 600;

      return SingleChildScrollView(
        child: Center(
          child: ConstrainedBox(
            constraints: const BoxConstraints(
              maxWidth: 1200,
            ),
            child: Padding(
              padding: EdgeInsets.all(isMobile ? 16 : 32),
              child: Column(
                crossAxisAlignment: CrossAxisAlignment.start,
                children: [
                  const Text(
                    'YOUR WORKSPACE',
                    style: TextStyle(
                      color: primaryColor,
                      fontSize: 12,
                      letterSpacing: 2,
                      fontWeight: FontWeight.bold,
                    ),
                  ),
                  const SizedBox(height: 10),
                  Text(
                    'Build something amazing.',
                    style: TextStyle(
```

```
    fontSize: isMobile ? 27 : 36,
    fontWeight: FontWeight.bold,
    color: const Color(0xFF252542),
  ),
),
const SizedBox(height: 8),
const Text(
  'Welcome to your Flutter web dashboard.',
  style: TextStyle(
    fontSize: 15,
    color: Color(0xFF77778C),
  ),
),
const SizedBox(height: 28),

// Welcome banner
Container(
  width: double.infinity,
  padding: EdgeInsets.all(
  isMobile ? 24 : 36,
  ),
  decoration: BoxDecoration(
    gradient: const LinearGradient(
    colors: [
        Color(0xFF6558E8),
        Color(0xFF9275F5),
      ],
      begin: Alignment.topLeft,
      end: Alignment.bottomRight,
    ),
    borderRadius: BorderRadius.circular(24),
  ),
  child: isMobile
      ? Column(
          crossAxisAlignment:
              CrossAxisAlignment.start,
          children: [
            _welcomeContent(context),
            const SizedBox(height: 24),
            const Icon(
            Icons.rocket_launch_rounded,
            color: Colors.white70,
              size: 75,
            ),
          ],
        )
      : Row(
          mainAxisAlignment:
              MainAxisAlignment.spaceBetween,
```

```
            children: [
              Expanded(
                child: _welcomeContent(context),
              ),
              const Icon(
                Icons.rocket_launch_rounded,
                color: Colors.white70,
                size: 100,
              ),
            ],
          ),
        ),

        const SizedBox(height: 32),
        const Text(
          'Explore Features',
          style: TextStyle(
          fontSize: 23,
            fontWeight: FontWeight.bold,
            color: Color(0xFF252542),
          ),
        ),
        const SizedBox(height: 16),

        // Responsive feature cards
        Wrap(
          spacing: 16,
          runSpacing: 16,
          children: [
            _featureCard(
              context,
              Icons.devices_rounded,
              'Responsive UI',
              'Adapts to every screen size.',
              const Color(0xFFEAE7FF),
            ),
            _featureCard(
              context,
              Icons.speed_rounded,
              'Fast Performance',
              'Smooth Flutter experience.',
              const Color(0xFFE0F7EF),
            ),
            _featureCard(
              context,
              Icons.touch_app_rounded,
              'Interactive',
              'Click and explore features.',
              const Color(0xFFFFEBD9),
```

```
                ),
              ].map((card) {
                return SizedBox(
                width: isMobile
                    ? double.infinity
                    : (constraints.maxWidth > 900
                        ? 350
                        : (constraints.maxWidth - 96) / 2),
                  child: card,
                );
              }).toList(),
            ),

            const SizedBox(height: 28),
            Center(
              child: Text(
                'Built with Flutter • Web Application',
                textAlign: TextAlign.center,
                style: TextStyle(
                  color: Colors.grey.shade600,
                  fontSize: 13,
                ),
              ),
            ),
          ],
        ),
      ),
    ),
  );
}

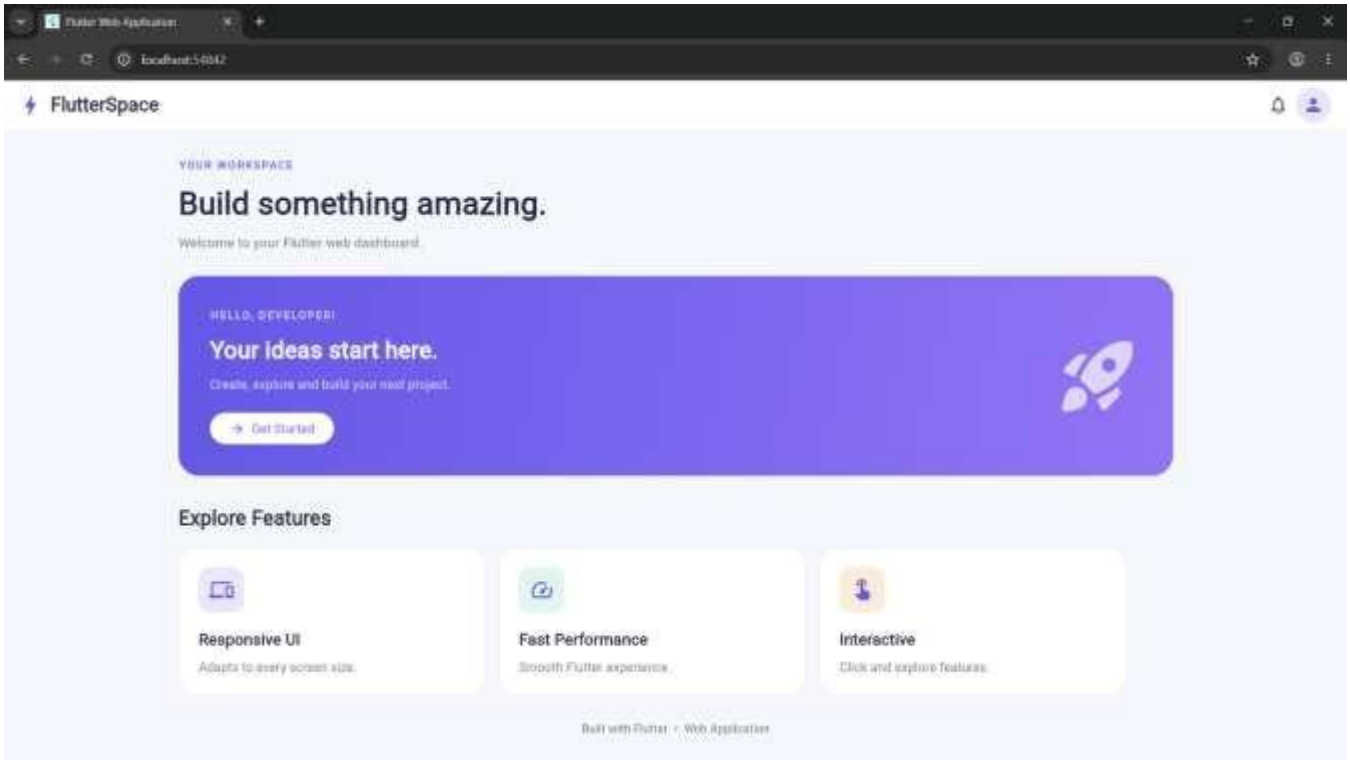
Widget _welcomeContent(BuildContext context) {
  return Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [
      const Text(
        'HELLO, DEVELOPER!',
        style: TextStyle(
          color: Colors.white70,
          fontSize: 12,
          letterSpacing: 2,
          fontWeight: FontWeight.bold,
        ),
      ),
      const SizedBox(height: 12),
```

```
      const Text(
        'Your ideas start here.',
        style: TextStyle(
        color: Colors.white,
        fontSize: 27,
          fontWeight: FontWeight.bold,
        ),
      ),
      const SizedBox(height: 10),
      const Text(
        'Create, explore and build your next project.',
        style: TextStyle(
          color: Colors.white70,
          fontSize: 14,
        ),
      ),
      const SizedBox(height: 22),
      ElevatedButton.icon(
      onPressed: () {
          ScaffoldMessenger.of(context).showSnackBar(
            const SnackBar(
              content: Text('Welcome! Your workspace is ready.'),
            ),
          );
        },
        icon: const Icon(Icons.arrow_forward_rounded),
        label: const Text('Get Started'),
        style: ElevatedButton.styleFrom(
          backgroundColor: Colors.white,
          foregroundColor: primaryColor,
          padding: const EdgeInsets.symmetric(
          horizontal: 22,
            vertical: 16,
          ),
        ),
      ),
    ],
  );
}

Widget _featureCard(
  BuildContext context,
  IconData icon,
  String title,
  String description,
  Color iconBackground,
) {
  return Container(
    padding: const EdgeInsets.all(22),
```

```
    decoration: BoxDecoration(
      color: Colors.white,
      borderRadius: BorderRadius.circular(20),
      border: Border.all(color: const Color(0xFFEEEEF5)),
    ),
    child: Column(
      crossAxisAlignment: CrossAxisAlignment.start,
      children: [
        Container(
          padding: const EdgeInsets.all(12),
          decoration: BoxDecoration(
          color: iconBackground,
            borderRadius: BorderRadius.circular(14),
          ),
          child: Icon(
            icon,
            color: primaryColor,
            size: 28,
          ),
        ),
        const SizedBox(height: 18),
        Text(
          title,
          style: const TextStyle(
            fontWeight: FontWeight.bold,
            fontSize: 18,
            color: Color(0xFF252542),
          ),
        ),
        const SizedBox(height: 8),
        Text(
          description,
          style: const TextStyle(
            fontSize: 14,
            color: Color(0xFF77778C),
          ),
        ),
      ],
    ),
  );
}
```

Output:



RESULT

A responsive Flutter Web application was successfully developed, executed in Chrome, tested with different screen sizes, and built for production.